Math 105LA Assignment 6

Fall 2019

Due Sun 11/17, 11pm

Write a MATLAB function to implement the modified Richardson iteration to solve the linear system $A\boldsymbol{x} = \boldsymbol{b}$. Recall that for vector iteration $\boldsymbol{x}^{(k)} = T\boldsymbol{x}^{(k-1)} + \boldsymbol{c}$ to converge, the spectral radius $\rho(T)$ must be less than 1. Your program should check if $\rho(T)$ is smaller 1 for $T = \boldsymbol{I} - \boldsymbol{wA}$.

The algorithm for modified Richardson iteration is

```
Input:  a matrix, A, a vector b, weight w, initial guess x0, tolerance tol,
        tolerance for the residual tol_r, maximum number of iteration N
Output: the approximate solution x
Step 1: Set n = 1
        Set T = I-w*A
        Set rho to be the eigenvalue of T with largest magnitude
Step 2: If |rho| >= 1
            Output('The iteration diverge'), stop
Step 3: While n <= N, repeat step 4 to step 8
Step 4:     Set v = A*x0-b
Step 5:     If ||v||_2 < tol_r
                  Output('||Ax-b|| is smaller than tol_r', x0), stop
Step 6:     Set x = x0-w*v;
Step 7:     If ||x-x0||_2 < tol
                  x0 = x
                  Output('||x_{n+1}- x_{n}|| is smaller than tol', x0), stop
Step 8:     Set x0 = x;
            Set n = n+1
Step 9: Output(x0), stop
```

You can start with the template below. Make sure that you use the **same order** of input:

```
function x = richardson(A,b,w,x0,N,tol,tol_r)
% A,b = the linear system A*x=b to be solved
% x0 = initial guess
% tol = tolerance
% tol_r = tolerance for the residual
% N = maximum number of iterations


end
```

Useful commands in MATLAB:

Find the eigenvalues of square matrix `A`	`eig(A)`
Find the norm of vector `x`	`norm(x)`

For example, suppose that we would like to solve linear system

$$\begin{cases} 2x_1 - x_2 + x_3 = -1 \\ -x_1 + 2x_2 - 2x_3 = 4 \\ x_1 - 2x_2 + 3x_3 = -5 \end{cases}$$

within 10^{-6} accuracy, using w=1/6, initial guess $\boldsymbol{x}_0 = [0,0,0]^T$ and at most 300 iterations. First define the coefficient matrix, right-hand side vector and the initial guess:

```
A = [2 -1 1; -1 2 -2; 1 -2 3];
b = [1; 4; -5];
w = 1/6;
x0 = [0; 0; 0];
```

Then, assuming your function is saved in '`richardson.m`', type the following in the command window:

```
x = richardson(A,b,w,x0,300,1e-6,1e-6)
```

And your function should return

```
x =
    2.0000
    2.0000
   -1.0000
```

More examples:

Define …	Then call your algorithm …	And it should return …

`A = [2 -1 1;`
`     -1 2 -2;`
`     1 -2 3];`
`W = 1;`
`b = [-1; 4; -5];`
`x0 = [0; 0; 0];`
	`richardson(A,b,w,x0,300,1e-6,1e-6)`	`The iteration diverge`

`A = [10 5 0 0;`
`      5 10 -4 0;`
`      0 -4 8 -1;`
`      0 0 -1 5];`
`w = 1/max(eig(A));`
`b = [6; 25; -11; 11];`
`x0 = [0; 0; 0; 0];`
	`richardson(A,b,w,x0,300,1e-6,1e-6)`	`||x_{n+1}-x_{n}|||| is smaller than 1.00e-06`
		`ans =`
		`-1.0047`
		` 3.2094`
		` 0.5176`
		` 2.3035`

`A = [10 5 0 0;`
`      5 10 -4 0;`
`      0 -4 8 -1;`
`      0 0 -1 5];`
`w = 1/(max(eig(A))+min(eig(A)));`
`b = [6; 25; -11; 11];`
`x0 = [0; 0; 0; 0];`
	`richardson(A,b,w ,x0,50, 1e-6,1e-6)`	`ans =`
		`   -1.0044`
		`    3.2089`
		`    0.5172`
		`    2.3033`

Submit a single file named by your UCInetID (e.g. `wtleung.m`) that contains the function. Your code will be tested on a few problems. Your grade of this assignment is calculated by the number of problems that your code solves correctly. Make sure your function runs without errors and has the same order of input above. **Failure to do so will result in a zero grade.**